

Deep Learning

Week 10

Question: 1

You are designing a surveillance system. You train your CNN to detect objects. You find that increasing the number of filters in the first layer from 8 to 64 improves accuracy but increases memory usage.

What might be the reason for this behavior?

Options:

- A. More filters capture diverse low-level features
- B. Increasing filters makes the model overfit and underperform
- C. Higher number of filters leads to higher feature map dimensionality
- D. First-layer filters determine high-level semantics

Correct Answer: A, C

Question: 2

Consider an input sequence $x = [2, 4, 6, 8]$ and a filter $w = [1, 0, -1]$. Using valid padding (no padding), what is the output of the convolution?

Options:

- A. $[-4, -4]$
- B. $[4, 4]$
- C. $[-4, 0, 4]$
- D. $[-2, -2]$

Correct Answer: A

Question: 3

You are processing satellite images to detect roads. You want to enhance edges and suppress smooth regions. Which of the following filters would be appropriate?

Options:

- A. $\begin{bmatrix} 1 & 1 & 1 \\ 1 & -8 & 1 \\ 1 & 1 & 1 \end{bmatrix}$
- B. $\begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}$

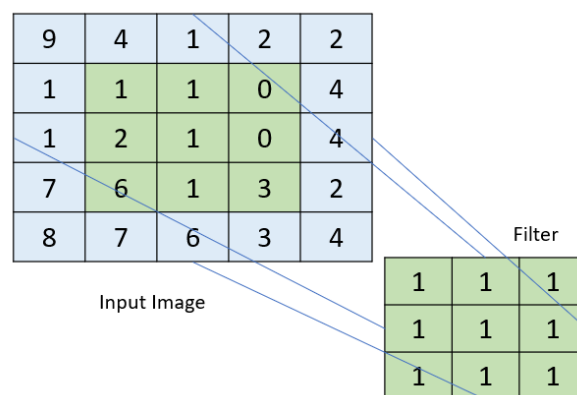
C. $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

D. $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Correct Answer: A, B

Explanation: A is a Laplacian filter (edge detection), B is a sharpening filter. C is blurring. D does nothing.

Question: 4



If a 3×3 filter with all weights 1 is applied over a 5×5 image, what effect would the filter have?

Options:

- A. Sharpen the image
- B. Blur the image
- C. Enhance edges
- D. Detect vertical lines

Correct Answer: B

Explanation: Uniform averaging blurs an image.

Question: 5

You are using a CNN layer with 8 filters, each of size 3×3 , on an RGB image. Which of the following statements are true?

Options:

- A. The filter depth is 3
- B. Each filter outputs a 2D feature map
- C. The output has 8 channels
- D. The total output shape is $H \times W \times 3$

Correct Answer: A, B, C

Explanation: Each $3 \times 3 \times 3$ filter processes across RGB. Each produces one 2D map. 8 filters produce 8 maps (channels). Option D is incorrect (output depth is 8, not 3).

Question: 6

If we apply multiple filters on an image input, what are the potential benefits?

Options:

- A. Each filter can learn to detect different features
- B. Feature maps capture spatial hierarchies
- C. It reduces the model parameters
- D. It improves translation invariance by itself

Correct Answer: A, B

Explanation: Filters extract different features (e.g., edges, corners). Translation invariance is aided by pooling, not filters alone.

Question: 7

You apply a 5×5 filter to a 32×32 input image using a stride of 1 and no padding. What will be the size of the output?

Options:

- A. 30×30
- B. 28×28
- C. 27×27
- D. 26×26

Correct Answer: B

Question: 8

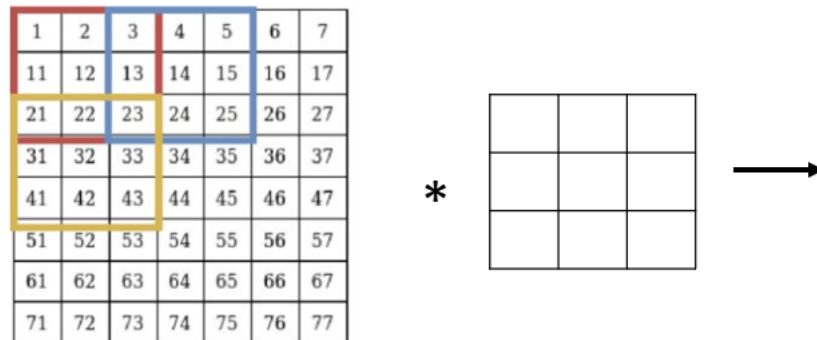
Which of the following are effects of increasing stride in a convolutional layer?

Options:

- A. Reduces output spatial dimensions
- B. Increases number of parameters
- C. Introduces more translation invariance
- D. Increases overlap between receptive fields

Correct Answer: A, C

Question: 9



You are designing a CNN layer with a 3×3 filter, stride 2, and no padding on a 7×7 input. What is the output dimension?

Options:

- A. 4×4
- B. 3×3
- C. 5×5
- D. 2×2

Correct Answer: B

Question: 10

What is the minimum amount of padding P required to preserve the input size after applying a 3×3 filter with stride = 1?

Options:

- A. 0
- B. 1
- C. 2
- D. 3

Correct Answer: B

Question: 11

Given an input of size 64×64 , a filter of size 5×5 is applied with a stride of 2 and padding of 1. What can you infer about the output?

Options:

- A. Output dimensions decrease
- B. Output shape is odd $\times$ odd
- C. Padding helps reduce dimensional loss
- D. Padding eliminates effect of stride

Correct Answer: A, B, C

Question: 12

A 128×128 grayscale image is processed using a convolutional layer with: Filter size = 7, Stride = 1, and Padding = 3

How many total pixels are in the output feature map?

Options:

- A. 16,384
- B. 14,884
- C. 15,876
- D. 15,376

Correct Answer: A

Question: 13

Which of the following are true about using stride > 1 and no padding?

Options:

- A. Encourages downsampling
- B. Increases output resolution
- C. Leads to loss of boundary information
- D. Makes convolution equivalent to pooling

Correct Answer: A, C

Question: 14

You apply two convolution layers back-to-back on an image of size 32×32 :

Conv1: 3×3 filter, stride=1, padding=1

Conv2: 3×3 filter, stride=2, padding=1

What is the final output size?

Options:

- A. 32×32
- B. 30×30
- C. 16×16
- D. 15×15

Correct Answer: C

Question: 15

You design a CNN and experiment with the following variations:

1. Using padding=0 vs padding=1
2. Using stride=1 vs stride=2
3. Changing filter size from 3×3 to 5×5

Which of the following are true impacts of these changes?

Options:

- A. Increasing padding helps retain more edge information
- B. Increasing stride helps reduce spatial size
- C. Larger filters increase the number of learnable parameters
- D. All three changes reduce the model's capacity

Correct Answer: A, B, C

Question: 16

What is the main purpose of pooling layers in a CNN?

Options:

- A. To increase feature map size
- B. To downsample the spatial dimensions and retain important information
- C. To normalize activations
- D. To eliminate overfitting

Correct Answer: B

Question: 17

Given a 4×4 feature map with values increasing from left to right and top to bottom, applying 2×2 max pooling with stride 2 gives an output of:

Options:

- A. 4×4
- B. 3×3
- C. 2×2
- D. 1×1

Correct Answer: C

Explanation:

$(4 - 2) / 2 + 1 = 2 \rightarrow$ pooled feature map is 2×2

Question: 18

Applying max pooling with a 2×2 filter and a stride of 1 on the following feature map produces:

$$\begin{bmatrix} 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 1 \\ 3 & 4 & 4 & 1 \\ 1 & 2 & 2 & 3 \end{bmatrix}$$

Options:

A) $\begin{bmatrix} 4 & 4 \\ 4 & 4 \end{bmatrix}$

B) $\begin{bmatrix} 4 & 4 & 4 \\ 4 & 4 & 4 \\ 4 & 4 & 4 \end{bmatrix}$

C) $\begin{bmatrix} 3 \\ 4 \\ 4 \\ 3 \end{bmatrix}$

D) $[3 \quad 4 \quad 4 \quad 3]$

Correct Answer: B

Question: 19

You are given the following 4×4 feature map:

$$\begin{bmatrix} 1 & 3 & 2 & 4 \\ 5 & 6 & 1 & 2 \\ 9 & 8 & 3 & 1 \\ 4 & 7 & 2 & 0 \end{bmatrix}$$

You apply both Max Pooling and Average Pooling using:

- Filter size = 2×2
- Stride = 2
- No padding

Which of the following are correct?

Options:

- A. The output of both max and average pooling will be of size 2×2
- B. The max pooled output contains the value 2
- C. The average pooled output has a higher total sum than the max pooled output
- D. Max pooling retains more spatial information than average pooling

Correct Answers: A, B

Question: 20

Which of the following best explains why pooling introduces translation invariance in CNNs?

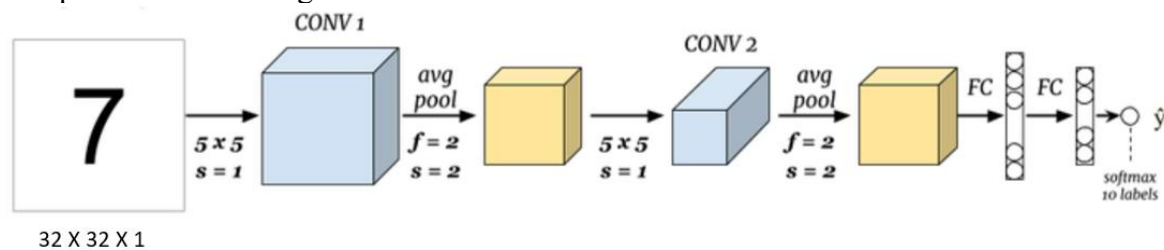
Options:

- A. Pooling layers apply filters across multiple channels
- B. Pooling selects dominant patterns over a region, making exact position less relevant
- C. Pooling reduces the model's capacity
- D. Pooling increases the filter size during convolution

Correct Answer: B

Question: 21

What is the output size of the feature map after the second pooling layer in LeNet, assuming the input is a 32×32 image?



Options:

- A. 6×6
- B. 10×10
- C. 8×8
- D. 5×5

Correct Answer: D

Question: 22

Which of the following were innovations introduced in AlexNet that contributed to its success on ImageNet?

Options:

- A. ReLU activation instead of sigmoid
- B. Training on multiple GPUs
- C. Use of depthwise separable convolutions
- D. Data augmentation and dropout

Correct Answer: A, B, D

Question: 23

What was the primary design principle behind VGGNet that enabled it to go deeper than earlier models?

Options:

- A. Use of large filters (11×11 and 7×7)
- B. Replacing large filters with small 3×3 filters stacked deeper
- C. Removing pooling layers entirely
- D. Using only fully connected layers

Correct Answer: B

Question: 24

Which of the following are true design choices in GoogLeNet (Inception) architecture?

Options:

- A. Use of 1×1 convolutions for dimensionality reduction
- B. Concatenation of outputs from multiple filter sizes
- C. Replacing all fully connected layers with global average pooling
- D. Use of skip connections between all layers

Correct Answers: A, B, C

Question: 25

Given an input feature map of shape $W \times H \times D$, applying a 1×1 convolution with K filters, stride = 1, and no padding, will result in an output of shape:

Options:

- A. $(W - 1) \times (H - 1) \times K$
- B. $W \times H \times D$
- C. $W \times H \times K$
- D. $K \times K \times D$

Correct Answer: C

Question: 26

You try to train a 50-layer CNN without skip connections, and the training accuracy saturates very early. Then you insert skip connections and observe immediate improvement. Which of the following best explains the improvement?

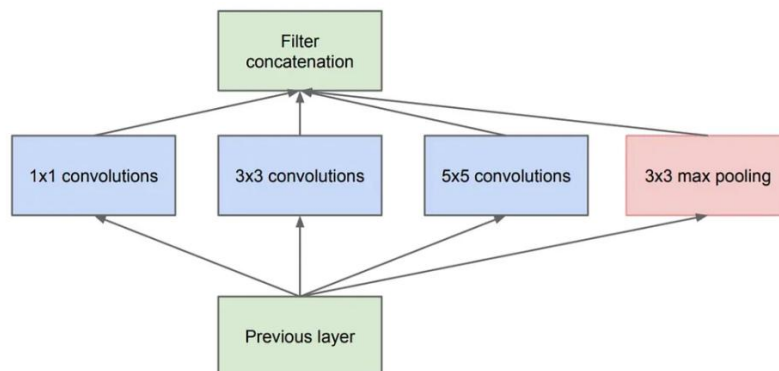
Options:

- A. The skip connections reduce overfitting
- B. The skip connections increase parameter count
- C. The skip connections help mitigate vanishing gradients
- D. The skip connections replace convolution layers

Correct Answer: C

Question: 27

What is the primary advantage of using parallel filter branches (1×1 , 3×3 , 5×5) in Inception modules?



Options:

- A. Reduced model size
- B. Ability to capture features at multiple scales
- C. Only works for grayscale images
- D. Forces sparsity in weights

Correct Answer: B

Explanation:

Parallel convolutions allow the model to learn both fine- and coarse-level features in the same layer.

Question: 28

You compare five CNN architectures trained on the same dataset:

- Model A: LeNet
- Model B: AlexNet
- Model C: VGGNet
- Model D: GoogLeNet
- Model E: ResNet

Which of the following architectural characteristics show a general trend or notable change across models A to E?

Options:

- A. Increasing network depth
- B. Increasing number of learnable parameters
- C. Introduction of residual (skip) connections
- D. Reduction in use of fully connected layers

Correct Answers: A, B, C, D

Question: 29

Which architecture has the highest number of parameters among LeNet, AlexNet, VGG-16, and ResNet-50?

Options:

- A. LeNet
- B. AlexNet
- C. VGG-16
- D. ResNet-50

Correct Answer: C

Question: 30

You apply a 1D convolution on the sequence $x = [1, 2, 3, 4, 5]$ with filter $w = [1, 0, -1]$, using $\text{stride} = 1$ and no padding.

What is the resulting output?

Options:

- A. $[-2, -2, -2]$
- B. $[2, 2, 2]$
- C. $[-2, 0, 2]$
- D. $[0, 0, 0]$

Correct Answer: A